

Digital Logic Circuits (Spring 2026)

HW #5

4. For the ripple counter in Figure 9–66, show the complete timing diagram for sixteen clock pulses. Show the clock, Q_0 , Q_1 , and Q_2 waveforms.

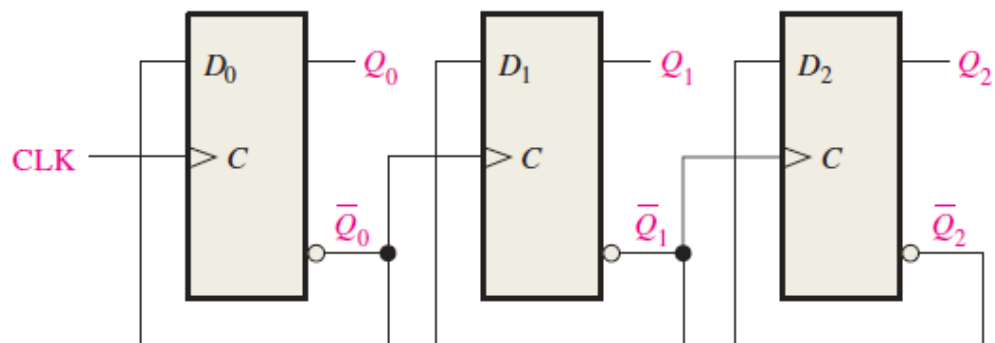


FIGURE 9–66

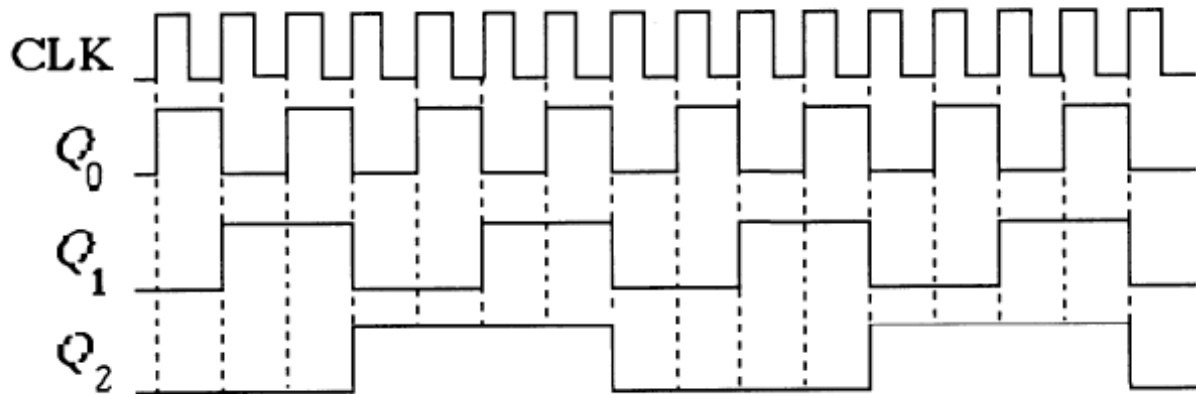


FIGURE 9-3

5. In the counter of Problem 4, assume that each flip-flop has a propagation delay from the triggering edge of the clock to a change in the Q output of 8 ns. Determine the worst-case (longest) delay time from a clock pulse to the arrival of the counter in a given state. Specify the state or states for which this worst-case delay occurs.

18. Determine the sequence of the counter in Figure 9–73.

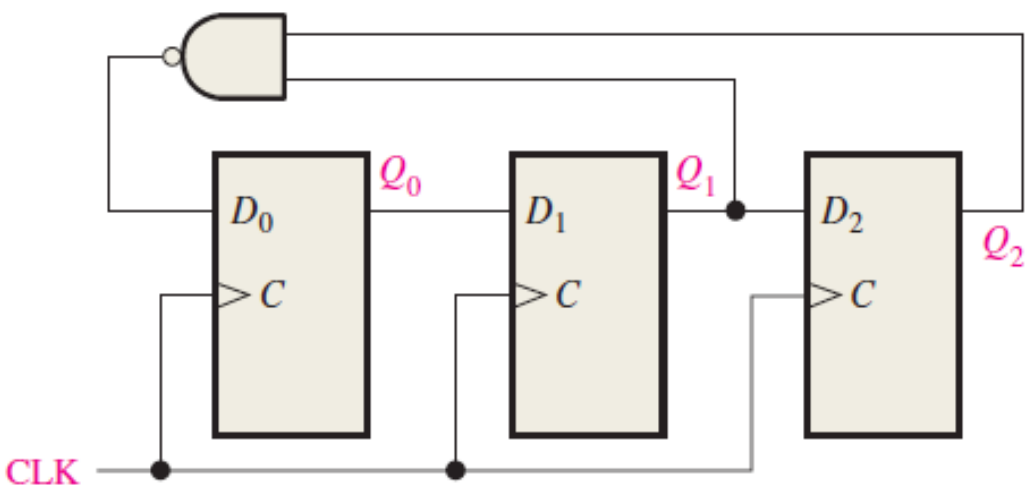


FIGURE 9–73

	Q_2	Q_1	Q_0	D_2	D_1	D_0
Initially	0	0	0	0	0	1
At CLK 1	0	0	1	0	1	1
At CLK 2	0	1	1	1	1	1
At CLK 3	1	1	1	1	1	0
At CLK 4	1	1	0	1	0	0
At CLK 5	1	0	0	0	0	1
At CLK 6	0	0	1	0	1	1

The sequence is 000 to 001 to 011 to 111 to 110 to 100 and back to 001, etc.

21. Design a counter to produce the following binary sequence. Use J-K flip-flops.
1, 4, 3, 5, 7, 6, 2, 1, . . .

NEXT-STATE TABLE

Present State			Next State		
Q_2	Q_1	Q_0	Q_2	Q_1	Q_0
0	0	1	1	0	0
1	0	0	0	1	1
0	1	1	1	0	1
1	0	1	1	1	1
1	1	1	1	1	0
1	1	0	0	1	0
0	1	0	0	0	1

TRANSITION TABLE

Output State Transitions (Present state to next state)			Flip-flop Inputs					
Q_2	Q_1	Q_0	J_2	K_2	J_1	K_1	J_0	K_0
0 to 1	0 to 0	1 to 0	1	X	0	X	X	1
1 to 0	0 to 1	0 to 1	X	1	1	X	1	X
0 to 1	1 to 0	1 to 1	1	X	X	1	X	0
1 to 1	0 to 1	1 to 1	X	0	1	X	X	0
1 to 1	1 to 1	1 to 0	X	0	X	0	X	1
0 to 0	1 to 0	0 to 1	0	X	X	1	1	X
1 to 0	1 to 1	0 to 0	X	1	X	0	0	X

See Figure 9-15.

$Q_2 Q_1$		Q_0	
		0	1
00		X	1
01		0	1
11		X	X
10		X	X

$J_2 = Q_0$

$Q_2 Q_1$		Q_0	
		0	1
00		X	0
01		X	X
11		X	X
10		1	1

$J_1 = Q_2$

$Q_2 Q_1$		Q_0	
		0	1
00		X	X
01		1	X
11		0	X
10		1	X

$J_0 = \bar{Q}_1 + \bar{Q}_2$

$Q_2 Q_1$		Q_0	
		0	1
00		X	X
01		X	X
11		1	0
10		1	0

$K_2 = \bar{Q}_0$

$Q_2 Q_1$		Q_0	
		0	1
00		X	X
01		1	1
11		0	0
10		X	X

$K_1 = \bar{Q}_2$

$Q_2 Q_1$		Q_0	
		0	1
00		X	1
01		X	0
11		X	1
10		X	0

$K_0 = \bar{Q}_1 \bar{Q}_2 + Q_1 Q_2$

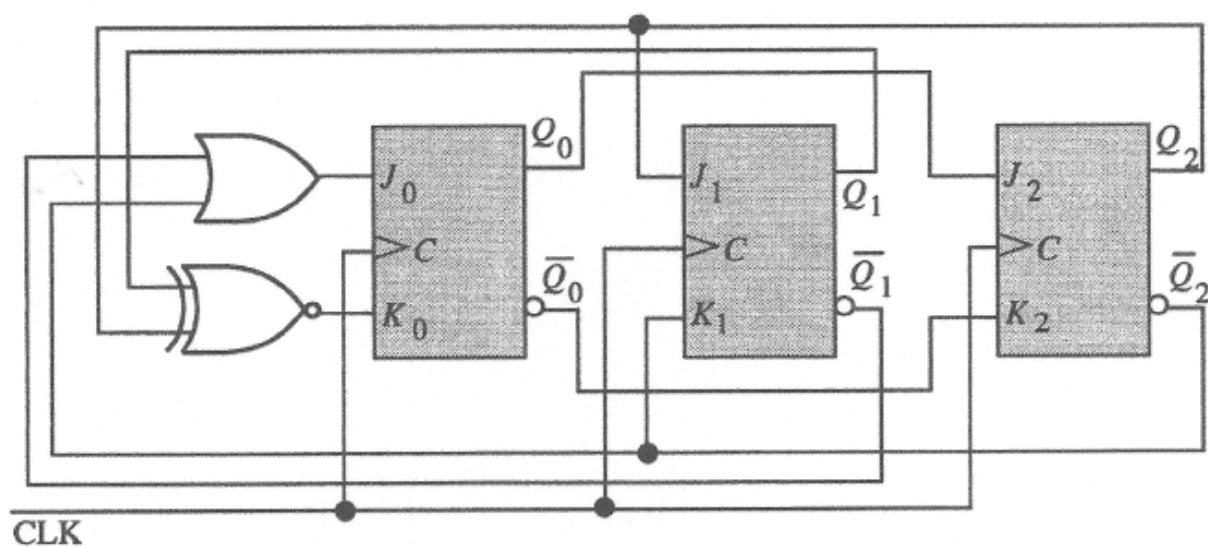


FIGURE 9-15

9. Determine the output of the array in Figure 10-67. The Xs represent connected links.

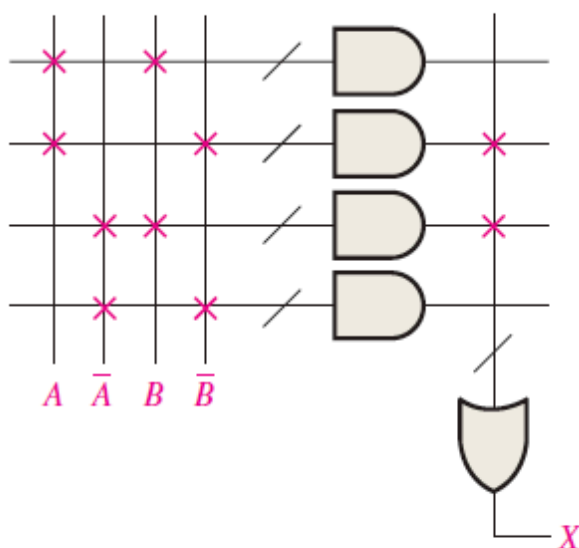


FIGURE 10-67

$$\overline{A}B + A\overline{B}$$

10. Modify the array in Figure 10–67 to produce an output $X = \overline{A}\overline{B}\overline{C} + \overline{A}B\overline{C} + ABC + A\overline{B}C$

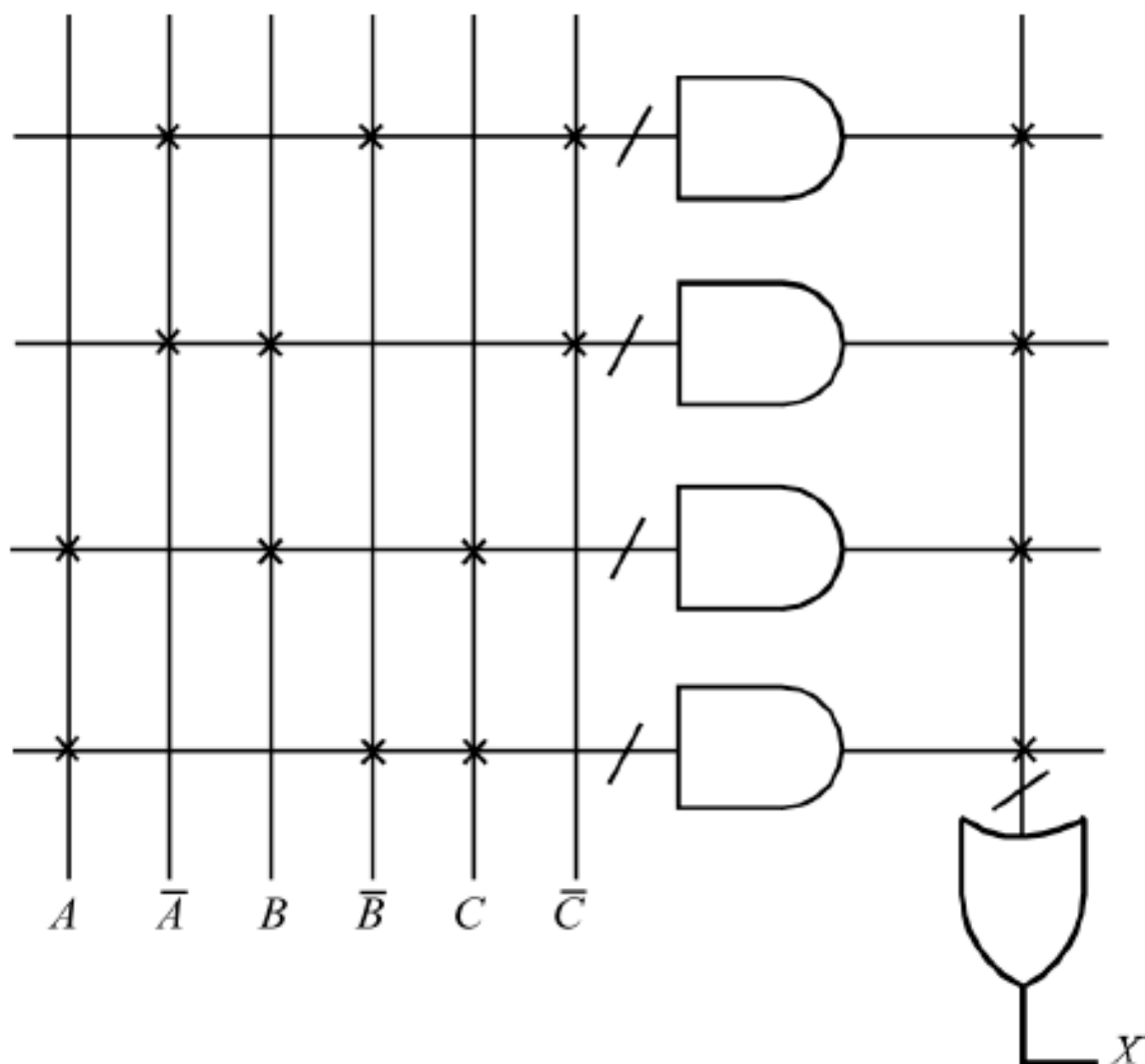


FIGURE 10-4

16. Generally, what elements make up a configurable logic block (CLB) in an FPGA? What elements make up a logic module?

An FPGA typically consists of configurable logic blocks (CLBs). Each CLB is made up of a number of logic modules with a local interconnect. Each logic module typically consists of a look-up table (LUT) and associated logic. Global column and row interconnects are used to connect the CLBs to I/Os as well as each other.

12. Determine the truth table for the ROM in Figure 11–80.

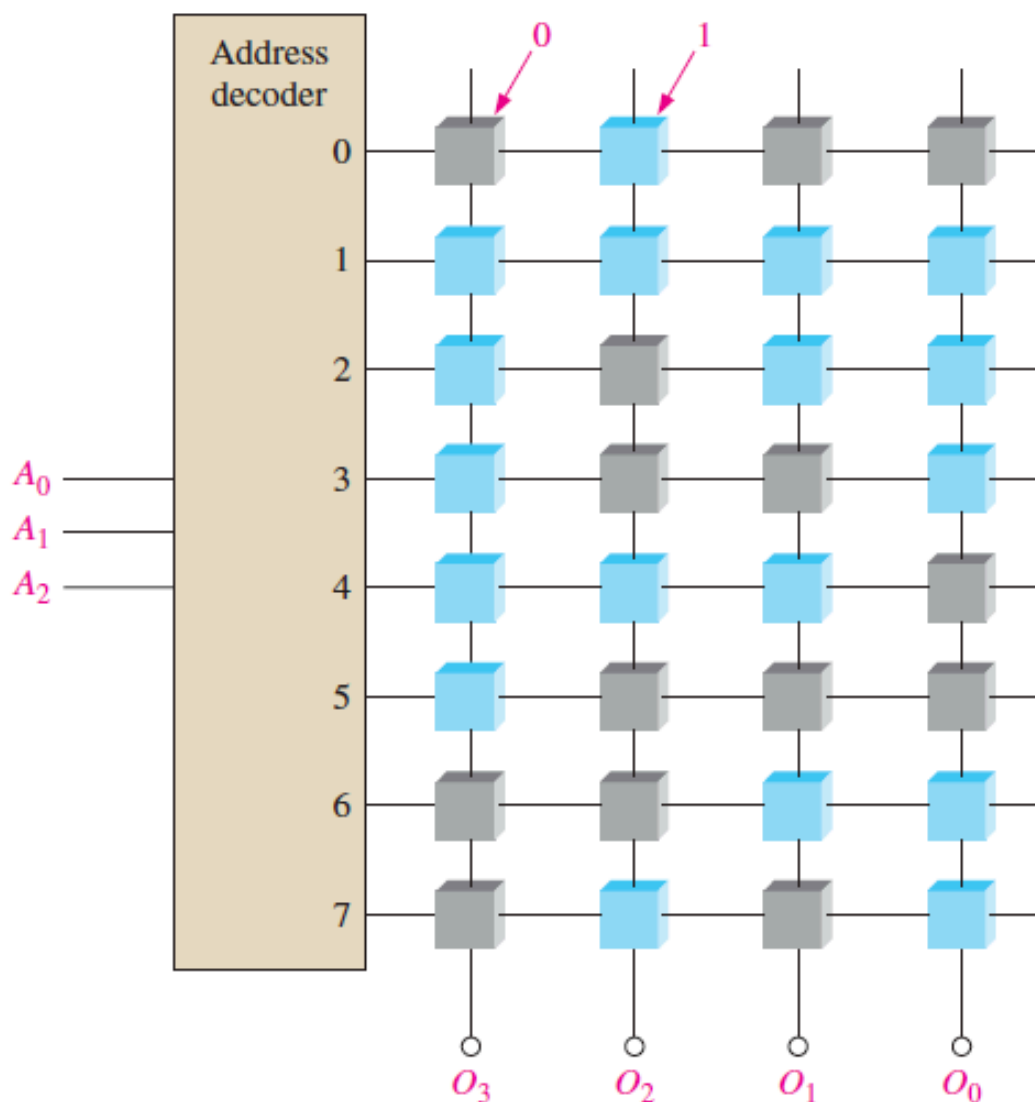


FIGURE 11–80

Inputs			Outputs			
A_2	A_1	A_0	O_3	O_2	O_1	O_0
0	0	0	0	1	0	0
0	0	1	1	1	1	1
0	1	0	1	0	1	1
0	1	1	1	0	0	1
1	0	0	1	1	1	0
1	0	1	1	0	0	0
1	1	0	0	0	1	1
1	1	1	0	1	0	1

17. Use $16k \times 4$ DRAMs to build a $64k \times 8$ DRAM. Show the logic diagram.

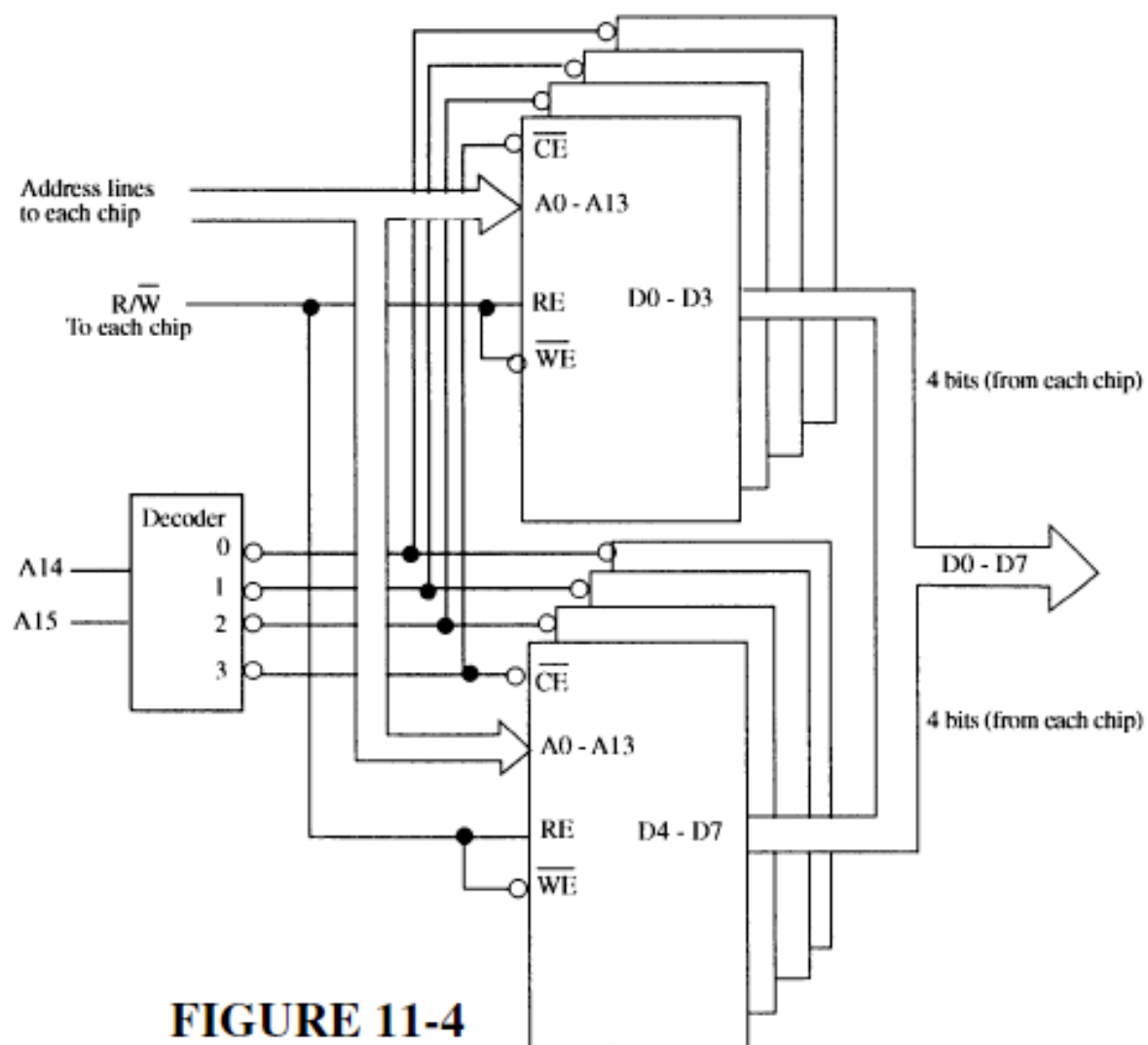


FIGURE 11-4